

HW2 Math 638 Differential Geometry II, Winter 2026

Due on class in paper, Wednesday January 21, 2026.

Q2.1. ([Lee, 2nd, 7-16 and 7-23, p.172–173]) (1a) Prove that $SU(2)$ is diffeomorphic to S^3 .

(1b) Let H^* be the Lie group of nonzero quaternions, and let $\mathcal{S} \subset H^*$ be the set of unit quaternions. Show that $\mathcal{S}$ is a properly embedded Lie subgroup of H^* , isomorphic to $SU(2)$.

Q2.2. ([Lee, 2nd, 7-19, p.172]) Suppose G , N , and H are Lie groups. Prove that G is isomorphic to a semidirect product $N \rtimes H$ if and only if there are Lie group homomorphisms $\varphi : G \rightarrow H$ and $\psi : H \rightarrow G$ such that $\varphi \circ \psi = \text{Id}_H$ and $\text{Ker } \varphi = N$.

Q2.3. Show that $\mathbb{R}^3$ with the bilinear operation of cross product is a Lie algebra.

Q2.4. Let G be a Lie group and let $\varphi : G \rightarrow G$ defined by $\varphi(g) = g^{-1}$. Prove

(4a) If X is a left invariant vector field, then $d\varphi(X)$ is a right invariant vector field with $d\varphi(X)(e) = -X(e)$.

(4b) Let $\mathfrak{g}_L$ and $\mathfrak{g}_R$ be the Lie algebra of G defined by left and right invariant vector fields, respectively. Show that $d\varphi : \mathfrak{g}_L \rightarrow \mathfrak{g}_R$ is a Lie algebra isomorphism.

Q2.5. Prove that $U(n)$ is diffeomorphic to $S^1 \times SU(n)$.

Q2.6. (6a) Prove that every matrix $A \in GL(n; \mathbb{C})$ can be uniquely written as $A = PR$, where P is positive definite hermitian matrix and R is a unitary matrix.

(6b) Prove that $GL(n; \mathbb{C})$ is connected.

(Hint: we assume the exponential map of matrix $B \rightarrow e^{iB}$ from hermitian matrices to unitary matrices is onto map.)

Q2.7. (7a) Let G be a connected Lie group and let $\varphi : G \rightarrow H$ be a Lie group homomorphism with discrete kernel. Prove that the kernel lies in the center of G .

(7b) Use (7a) to prove that the fundamental group of a Lie group is abelian.